

Anoop Kamlakar Hattekar

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SUMMARY

Embedded systems enthusiast with strong foundation in firmware development using **C/C++** on **ESP32**, **STM32**, **Arduino**, **PIC18F**, and **8051** platforms. Experienced in peripheral interfacing, real-time control, and hardware-level debugging. Passionate about hardware-software integration with strong electronics fundamentals and genuine curiosity for solving embedded challenges.

EXPERIENCE

TechB Softwares

February 2026 – Present

Full Stack Developer Intern

Pune, India

- Worked on backend logic and data processing in **Python/Django** for energy-based applications involving real-time data handling
- Designed and integrated **REST APIs** ensuring reliable communication between frontend and backend systems; used **Git** for version control and code reviews
- Gained exposure to scalable system design, debugging, and handling production-level codebases

TECHNICAL SKILLS

Programming: C, C++, Embedded C, Data Structures

MCUs (8/16/32-bit): ESP32 (32-bit), STM32 (32-bit), Arduino, 8051 (8-bit), PIC18F

Protocols: UART, USART, SPI, I2C, Bluetooth

Peripherals: ADC, PWM, Timers, GPIO

Test Equipment: MSO, DSO, Oscilloscope, Logic Analyzer, Multimeter, Serial Debugging

Tools: Keil uVision, STM32CubeIDE, Arduino IDE, Git, GitHub

OS: Linux (Ubuntu)

PROJECTS

Line Follower Pick & Drop Rescue Bot | ESP32, Raspberry Pi, Embedded C, PID, UART, USART, I2C

2026

- Designed and developed an autonomous rescue bot using **ESP32** as the main controller and **Raspberry Pi** for real-time camera processing; implemented **PID**-based control algorithm for precise line following and dynamic path correction
- Implemented **Bluetooth** wireless communication between ESP32 and control interface; integrated multiple peripherals including **PWM** motor control, **ADC**, **GPIO**, and timers
- Applied **UART**, **USART**, and **I2C** protocols for inter-module data exchange; performed hardware-firmware validation via logic analyzer and oscilloscope-based signal verification

Smart Wheelchair System | Arduino Nano, Embedded C/C++, I2C, PWM, Interrupts

2024

- Developed interrupt-driven firmware in **Embedded C/C++** for motor control using **PWM** and real-time input handling; implemented built-in diagnostics for real-time fault detection
- Interfaced motor drivers and sensors using **I2C** communication; performed serial debugging and oscilloscope-based signal verification; assembled and tested circuit boards
- Recognized with **Government of India Copyright** (Reg. No: LD-20250167659) for innovation in embedded assistive technology

ACHIEVEMENTS

- **Project Competition Winner (2024):** Selected for **SPPU State-level** competition among 100+ participants
- **Government Copyright (2025):** Reg. No: LD-20250167659 — Recognized for innovation in embedded assistive technology

EDUCATION

PCET's Pimpri Chinchwad College of Engineering and Research

Expected June 2027

Bachelor of Engineering — Electronics and Telecommunication

CGPA: 9.1 / 10.0

JK Junior College of Science

February 2023

Higher Secondary Certificate (HSC)

76.67%